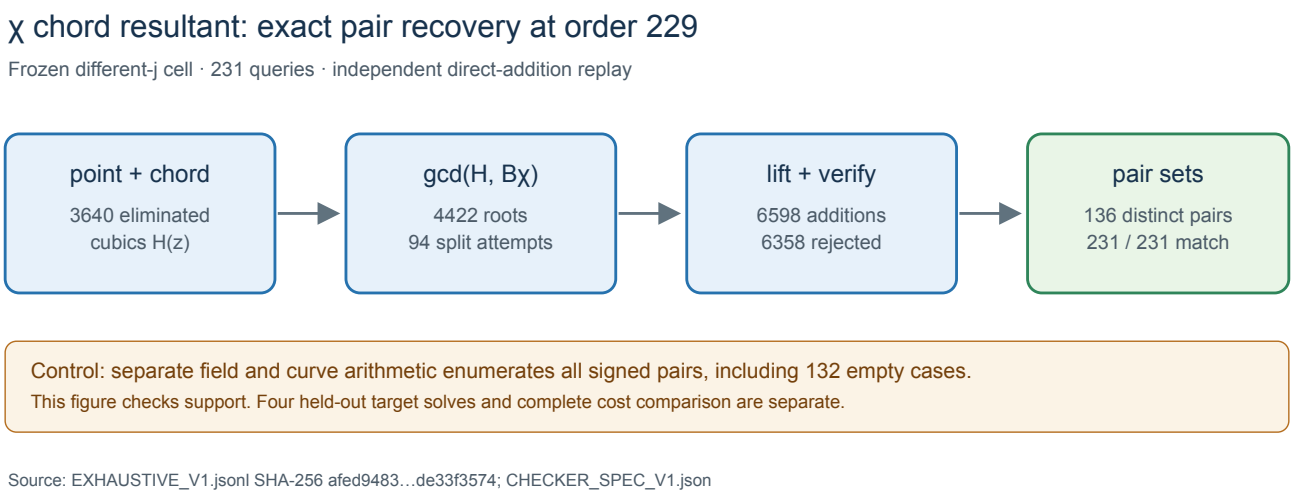
Direct χ chord resultant: exhaustive pair control

2026-10-16 · frozen different-j order-229 cell

The eliminated chord polynomial was intersected with the signed factor-base χ polynomial. All roots were lifted and checked by full source addition. A separate Rust checker rebuilt the curve and enumerated every pair with independent field arithmetic.



Source: EXHAUSTIVE_V1.jsonl SHA-256 afed9483…de33f3574; CHECKER_SPEC_V1.json

Across all 229 subgroup points and the two torsion exceptions, the returned unordered pair set matched direct enumeration in 231 of 231 cases. There were 136 distinct valid pairs and 132 cases with no decomposition. The oracle made 3640 resultants, 7410 gcds, 94 split attempts and 6598 exact candidate additions; 6358 candidates were rejected. The metered repeat used 7,094,272 bytes peak RSS, below the 512 MiB ceiling, and its pair rows matched the first run exactly.

The complete relation solve, portable primitive price and isolated wall comparison have not been measured by this exactness run. Sources: `RUN_SPEC_V1.json`, `EXHAUSTIVE_V1.jsonl`, `CHECKER_SPEC_V1.json`, `EXHAUSTIVE_V1.check.json`, `RESULTS_V1.md`.